



Review article

Multimodal artificial intelligence in ophthalmology: Applications, challenges, and future directions

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ABSTRACT

With the rapid development of artificial intelligence (AI) technology, multimodal AI that integrates multiple data modalities has shown tremendous potential in the field of ophthalmology. We systematically evaluate the current applications, technical characteristics, and clinical value of multimodal AI in ophthalmology. We conducted a systematic review following the PRISMA guidelines. We searched relevant literature published from 2018 to 2025 in PubMed, Web of Science, Scopus, and Google Scholar databases. A total of 10 studies were included in the final analysis. The main applications of multimodal AI in ophthalmology include glaucoma, age-related macular degeneration, corneal diseases, ophthalmic emergency triage, cognitive impairment screening, diabetes complication screening, and chatbot-based ophthalmic consultation. Multimodal systems showed superior performance compared to unimodal systems across various application areas, with an Area Under the Curve improvements of 4–5 % and accuracy improvements of 2–7 %. Multimodal AI demonstrates broad application prospects in ophthalmology, providing more comprehensive and accurate diagnostic information. Future research should focus on clinical validation, novel fusion methods, interpretability, and lightweight models to promote the clinical translation of multimodal AI in ophthalmology.

1. Introduction

Ophthalmology is a medical specialty that highly depends on various examination methods, often requiring the integration of different modalities of data for diagnosis and treatment decisions in clinical practice.¹¹ Traditional ophthalmic diagnosis mainly relies on physicians' experience and subjective judgment, which has certain limitations and variability. With the rapid development of artificial intelligence (AI) technology, especially breakthroughs in deep learning algorithms, AI systems based on single-modal data (such as fundus photographs or optical coherence tomography [OCT]) have achieved significant results in the diagnosis of ophthalmic diseases^{3,14,23}; however, single-modal AI

systems often cannot comprehensively capture the multidimensional features of diseases and struggle to simulate the diagnostic thinking of clinical physicians who integrate multiple sources of information.⁴⁰

Multimodal AI, by integrating data from different sources and of different types, can provide a more comprehensive representation of diseases, thereby achieving more accurate diagnosis and prediction.²⁵ In ophthalmology, multimodal data may include different types of imaging examinations (such as fundus photographs, OCT, OCT angiography [OCTA], etc.), functional examinations (such as visual field tests, electrophysiology, etc.), clinical data (such as medical history, symptom descriptions, etc.), and laboratory test results.³⁸ Common AI algorithms in multimodal systems include convolutional neural networks (CNNs),

Abbreviations: AI, Artificial Intelligence; AMD, Age-related Macular Degeneration; AUC, Area Under the Curve; CNN, Convolutional Neural Network; CNV, Choroidal Neovascularization; DCNN, Deep Convolutional Neural Network; EE, Eye Emergency; FLF, Feature-Level Fusion; GA, Geographic Atrophy; HD, High Definition; IR, Infrared Reflectance; NNV, Non-Neovascular; NV, Neovascular; OCT, Optical Coherence Tomography; OCTA, Optical Coherence Tomography Angiography; OR, Odds Ratio; PCV, Polypoidal Choroidal Vasculopathy.

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